

英検 2^級

過去問

長文問題集

②

収録内容

2010年度第3回～2013年度第3回

3

A

次の英文[A], [B]を読み, その文意にそって(26)から(33)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

From Factories to Farms

As the world's economy changes, new industrial cities develop. One result is that many older industrial cities find themselves losing the factories and jobs that made them wealthy. A good example of this is Detroit, in the United States. Detroit's modern history began at the end of the 19th century, when a man named Ransom Olds built a car factory there. The cars produced by Olds were very popular, and soon other car manufacturers opened factories in the city. (26), the population began to increase. By 1950, Detroit was the fourth-largest city in America.

Some time later, however, (27) for Detroit's cars began to drop. Other countries were producing cheaper and more efficient cars, and more and more of the city's factories went out of business. Now, Detroit's population has fallen by half, and the city is full of empty factories and wasteland. Unemployment and poverty are common, and people are searching for good ways to use the empty land. One idea that is becoming increasingly popular is to go back to how the land was used before the car factories came—farming.

This idea started as a way to help (28) people who needed food. Taja Sevelle, a singer, began creating small gardens on empty land and encouraging volunteers to grow fruit and vegetables. The food was then given away to anyone in the neighborhood who needed it. Her idea was very successful and has since spread to many other cities. Now, John Hantz, a wealthy businessman, is taking this idea a step further. He has begun buying up unused land in Detroit to use for commercial farming. He faces many problems in making his business work, including polluted land and high taxes, but he is confident that he will be able to overcome them. He believes that growing crops on the land will not only bring him a (29), but also help to make Detroit a much greener city.

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|------|--------------|-----------|----------------|------------------|
| (26) | 1 Naturally | 2 Rather | 3 Nevertheless | 4 Unfortunately |
| (27) | 1 permission | 2 demand | 3 assistance | 4 responsibility |
| (28) | 1 religious | 2 healthy | 3 young | 4 local |
| (29) | 1 decision | 2 task | 3 profit | 4 warning |

May I Take Your Order?

Many people have had the experience of visiting a restaurant where the waiter, instead of writing down customers' orders, simply remembers them. How do these waiters keep so much information in their heads? Customers are not the only people to (30) about this. Recently, scientists who study memory have also become interested in this question. Most experts believe that people use two types of memory—short-term memory and long-term memory. In general, people can only keep about four pieces of information in their short-term memories. Moreover, these memories fade after about 18 seconds. Many waiters, though, seem to have memories that perform much better.

Recently, a team of scientists in Buenos Aires, Argentina, carried out a study of waiters. The scientists wanted to do this because of the reputation of the city's waiters. Many of the older waiters in Buenos Aires's restaurants are (31) their ability to remember orders and deliver each one to the right customer without checking. The scientists had a group of eight volunteers visit restaurants and order drinks. The waiters remembered their orders and delivered each one to the right person, even when they had taken orders from other customers. However, after the volunteers ordered a second drink, they changed (32). Although the waiters were able to remember the orders, they became confused about which customer had ordered what drink. It was clear that the waiters were not remembering the customers but the places where they were sitting.

The scientists who carried out the study believe that the waiters create a map in their minds of where people are sitting. When the waiters take orders, they connect them to this map. This allows them to transfer information they hear into their long-term memories very quickly. The scientists hope that their research will help them develop ways of training that allow everybody to have memories as (33) as those of the waiters in Buenos Aires.

- | | | | | |
|------|----------------|-------------|--------------|--------------|
| (30) | 1 lie | 2 wonder | 3 protest | 4 reply |
| (31) | 1 silent about | 2 afraid of | 3 shocked by | 4 famous for |
| (32) | 1 clothes | 2 dates | 3 seats | 4 dishes |
| (33) | 1 accurate | 2 temporary | 3 varied | 4 official |

Genetic Clues

In recent years, it has become common for police to use DNA analysis to solve crimes. This usually involves a technique known as “genetic fingerprinting.” Most of the genetic information in DNA is the same in everyone, but a small percentage is unique to each individual. By comparing someone’s DNA to that found at the scene of a crime, it is possible to prove that he or she was there. Now, though, the police have begun expanding this DNA analysis to cats.

The usefulness of this was first shown in Canada. The police there suspected a man of carrying out a murder, but they had no evidence that he was present at the crime. They did, however, find a jacket nearby on which there were some white hairs. One of the policemen remembered that the parents of the man had a white cat. They asked a scientist to compare DNA from both the cat and the white hairs. They also took DNA samples from other cats in the neighborhood. The DNA from the hairs and the parents’ cat turned out to be the same, and different from that of the other cats. This showed that the jacket belonged to the man, and he was sent to prison for the crime.

The biggest difficulty with genetic fingerprinting is making sure which part of the DNA is unique. For this reason, it is important to have examples of the DNA of genetically related animals to identify which genes are shared and which are not. With this in mind, the United States Federal Bureau of Investigation (FBI) requested scientists to establish databases of animal DNA. In response, a team led by Robert Grahn, a scientist at the University of California at Davis, began to collect DNA from cats around the world. The team’s database now has DNA from 1,394 different cats.

Although the scientists plan to include the DNA of dogs and other animals, they believe that the cat DNA will be especially useful. This is because cats leave fur behind wherever they go. This makes it almost impossible for people nearby to avoid getting fur on their clothes. For this reason, police confidently expect the DNA of cats to play a part in many future crime investigations.

(37) The technique of “genetic fingerprinting”

- 1 was first tested on cats but is now also being used for human beings.
- 2 relies on the fact that the DNA of each individual is slightly different.
- 3 can be used to determine the types of people likely to commit crimes.
- 4 is a way of discovering similarities between two separate crime scenes.

(38) What did genetic fingerprinting show the police in Canada?

- 1 That the white hairs found on a jacket belonged to the victim of a murder.
- 2 That the jacket found at the crime scene belonged to the murderer’s father.
- 3 That a man they suspected of murder had been at the scene of the crime.
- 4 That a man had been wrongly sent to prison for murdering someone.

(39) What is one thing that the new database will do?

- 1 Show which genes are shared by genetically related animals.
- 2 Provide a list of cats that have been at crime scenes around the world.
- 3 Store information about experts on genetic fingerprinting.
- 4 Identify which types of DNA are unique to humans.

(40) Why do the police think cat DNA will be more useful to them than DNA from other animals?

- 1 Cats are less likely to be noticed at a crime scene.
- 2 Cats have a wider variety of DNA than other animals.
- 3 Cat fur tends to remain at a crime scene for a longer time.
- 4 Cat fur is more likely to be found on criminals’ clothes.

The Sky Is the Limit

When the United States joined World War II in 1941, the country found it necessary to send soldiers and supplies across the Atlantic Ocean to Europe. Many of the ships it used, however, were sunk by German submarines. An American shipbuilder named Henry J. Kaiser came up with the idea of building huge airplanes that could be used instead of ships to carry the troops and equipment. He contacted Howard Hughes, a millionaire and airplane designer, and suggested that they build such a plane together.

In 1942, Hughes and Kaiser signed a contract with the U.S. government to develop the Hercules, a giant plane that could take off from and land on the ocean. They promised that it would be large enough to carry 750 soldiers and their equipment. The major problem they faced, however, was a shortage of metals caused by the war. This meant that the plane had to be built almost entirely of wood. Work started at Hughes's company in California.

Although the plane was needed immediately, the work took far longer than expected. One reason was that even the small amount of metal they wanted was difficult to obtain. Another was Hughes's concern with unimportant details. He would, for example, spend days worrying about the design of the plane's control panel. Eventually, Kaiser left the project and Hughes carried on alone. By the time he finished, though, the war was over. The Hercules was three times larger than any airplane that had been built before. Hughes had spent \$22 million of the government's money and \$8 million of his own money building it.

Many people doubted that such a huge airplane could fly, and they criticized Hughes for spending so much public money on it. In 1947, Hughes was called to Washington, D.C., for questioning by the government. During the investigation's summer break, however, he returned to California to take the Hercules on a test flight. To everyone's surprise, the huge plane rose 21 meters above the water and flew for about 1.6 kilometers. Hughes had shown his critics that the airplane could fly, and the government ended the investigation. Although the Hercules never flew again, it became known as an engineering miracle, one that showed that a huge size does not make flying impossible.

(41) Why did Henry J. Kaiser want to build huge airplanes during World War II?

- 1** He hoped to sell them to Howard Hughes's airline company.
- 2** Ships carrying men and equipment were often attacked.
- 3** Such airplanes could be used to sink enemy submarines.
- 4** It was becoming expensive for him to make ships.

(42) What was one difficulty faced by Hughes and Kaiser?

- 1** There was a shortage of wood in California.
- 2** They could not get all the metals that they needed.
- 3** Hughes's factory was located far from the Atlantic Ocean.
- 4** The airplane could not be designed to take off from land.

(43) What was one reason the Hercules took so long to build?

- 1** Kaiser left the project because the war had ended.
- 2** It was difficult to find workers because of the war.
- 3** Hughes spent a lot of time thinking about minor details.
- 4** The government did not provide the money it had promised.

(44) Why did the government end its investigation of Hughes?

- 1** His airplane never flew again after its first flight across the sea.
- 2** The war finished long before his airplane was ready to be used.
- 3** A test flight showed that his airplane was able to fly despite its size.
- 4** He successfully flew his airplane from California to Washington, D.C.

(45) Which of the following statements is true?

- 1** After the war, many people said Hughes had wasted government money.
- 2** Kaiser and Hughes argued over how much money to spend on the plane.
- 3** The Hercules project was not completed because Hughes ran out of money.
- 4** Hughes went to Washington, D.C., to get back the money he had spent on the project.

3

A

次の英文 **A**, **B** を読み、その文意にそって (26) から (33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び、その番号を解答用紙の所定欄にマークしなさい。

A Useful Break

Recently in the United States, a growing number of students have begun taking a gap year—a year’s break between high school and college. This idea is a fairly new one in the U.S. In the United Kingdom and Australia, though, it has been (**26**) for some time. During this year, most students travel, work, or do volunteer activities. At first, many U.S. colleges were suspicious of gap years. They feared that students would (**27**) the habit of studying. Now, however, many colleges are recognizing that the experience can be valuable. In fact, they often find that it helps students become more enthusiastic about studying.

A good example is Linda Clark from Maryland, who decided to take a year off after high school. She spent the time working as a volunteer in Ghana and studying in Italy. While she was in Ghana, Clark taught math and English to local children. She also helped the community build a library for its school. After that, she went to Italy, where she studied art. According to Clark, Italy was fun, but Ghana changed her life. This was because her experiences there gave her a new (**28**). She decided she wanted to help countries like Ghana grow economically. She studied economics at college and now gives business advice to people in developing countries.

Many experts say that the most important thing is not to waste one’s time during a gap year. Many companies have begun helping students organize their gap year by arranging programs for them. However, British author Tom Griffiths warns that such expensive programs may not be the best way to spend the time. According to Griffiths, students do not mature unless they have to deal with situations on their own. “If you’re not in a group,” he says, “you get more out of it. It’s just a fact that if you travel around the world on your own, you (**29**).”

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|------|------------------|------------------|------------------|------------------|
| (26) | 1 uneasy | 2 illegal | 3 common | 4 private |
| (27) | 1 lose | 2 prefer | 3 control | 4 delay |
| (28) | 1 price | 2 aim | 3 hobby | 4 limit |
| (29) | 1 stop by | 2 run out | 3 give in | 4 grow up |

Patterns of Hope

One major problem for hospitals is the number of patients who catch dangerous diseases while they are being treated for other problems. These diseases are caused by the bacteria that live in hospitals and get into patients' bodies. The usual way to deal with these bacteria has been to use medicines or chemicals to try to kill them. Unfortunately, this method can have the opposite effect and actually cause the bacteria to become stronger. Now, Anthony Brennan, a scientist at the University of Florida, has come up with a new way to reduce bacteria in hospitals—one that could help avoid this (**30**).

Brennan first thought of his idea when he was working with the United States Navy. The Navy was looking for a way to prevent tiny animals and plants from attaching themselves to ships. Over time, a thick layer of them can develop, and this both damages the ships and reduces the speed at which they can travel. Brennan began thinking about how sea creatures deal with the same problem. Most of those that move slowly, such as whales and turtles, do in fact become covered by such a layer. Slow-moving sharks, though, have skin that remains (**31**). When Brennan examined these sharks closely, he discovered that this was because of patterns of tiny bumps on their skin. These patterns make it hard for the small creatures and plants to stick to the skin.

Brennan decided to create a (**32**) surface that could be used in hospitals. Eventually, he created a plastic layer on which patterns like those on shark skin are printed. Experiments have shown that bacteria find it very hard to live on these patterns. Brennan hopes that hospitals will use his plastic layer to cover surfaces where bacteria usually gather. If his idea succeeds, this will be a good example of how much we human beings can (**33**) from nature.

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|------|------------|------------|-----------|-----------|
| (30) | 1 attitude | 2 risk | 3 promise | 4 meaning |
| (31) | 1 dark | 2 flexible | 3 rough | 4 clean |
| (32) | 1 similar | 2 basic | 3 cheap | 4 liquid |
| (33) | 1 remove | 2 guard | 3 learn | 4 explain |

Robot Manners

As robot technology improves, robots are becoming an increasingly important part of our daily lives. Robots, of course, have many advantages over human beings. They work hard, never need to rest, do not complain, and do not need to be paid. Moreover, by performing simple tasks, they make it possible for human workers to focus on jobs that require more skills. Yet as more robots enter the workplace, their creators face a new problem—how to design them so that human beings can get along with them.

A typical robot today is TUG, which is now used in more than 50 hospitals in the United States. TUG is a box-like robot on wheels that is able to carry things, push carts, and deliver messages. TUG is programmed to remember all the places in a hospital. When a nurse wishes to send something to another nurse, for example, instead of having to waste time carrying it, he or she puts it on TUG, selects the destination, and the robot goes there on its own. Hospital managers like TUG because it allows hospital workers to concentrate on the jobs they were originally trained to do.

Unfortunately, not all hospital workers like TUG. Bilge Mutlu, a specialist in human-robot interaction at the University of Wisconsin–Madison, studied the reactions of workers to TUG in one hospital. He found that many people disliked the robot's lack of manners. For example, when a human being delivers a message to someone who is talking on the telephone, he or she will usually wait until the person on the phone is free. TUG simply interrupts the call and loudly announces, "TUG has arrived."

Aethon, the company that makes TUG, is taking Mutlu's research seriously. Aethon has already programmed TUG to speak more quietly at night. However, it is very difficult to program a robot to be polite. One possible answer, being developed by scientists at the University of Washington, in Seattle, is a program that allows robots to analyze and imitate human behavior. This will only work, however, if the human beings that the robots copy are themselves behaving politely.

- (37) What will robot designers need to do from now on?
- 1 Pay robots for performing simple tasks.
 - 2 Prevent robots from replacing human beings.
 - 3 Build robots that can concentrate on highly skilled jobs.
 - 4 Create robots that people feel comfortable working with.
-

- (38) Why do people who run hospitals like TUG?
- 1 It can guess what task it should do next.
 - 2 It can move easily from one hospital to another.
 - 3 It can stop workers from wasting time on simple jobs.
 - 4 It can help new nurses find their way around the hospital.
-

- (39) Bilge Mutlu discovered that people in one hospital
- 1 felt that the way TUG acted was impolite.
 - 2 disliked it when TUG came too close to them.
 - 3 refused to listen to TUG's messages.
 - 4 thought that TUG worked too hard.
-

- (40) What are some scientists trying to do?
- 1 Use robots to teach people how to be more polite to each other.
 - 2 Program robots to behave more like human beings.
 - 3 Analyze the reasons why robots sometimes stop working.
 - 4 Design robots that do more work at night than they do in the day.

Saving a Copy

In 1993, director Steven Spielberg had a big hit with his movie *Jurassic Park*. The movie was about a theme park where people could see cloned dinosaurs, but it was criticized by some scientists as unrealistic. Dinosaurs, they said, could never be brought back to life by cloning. What those scientists did not know, though, was how much progress cloning would make after the movie was produced. In particular, a new method known as “interspecies cloning” has brought the world of Spielberg’s movie much closer to reality.

Interspecies cloning means using the egg of one species of animal to produce a clone of another species. First, an egg is taken out of a female animal and the DNA is removed. Then the DNA of another animal is put into the egg. Electricity is passed through the egg, causing it to start growing. The egg is then put back into the mother’s body, where it develops into a normal baby. Scientists have so far managed to create a number of animal clones in this way.

One scientist who has been working on interspecies cloning is Betsy Dresser of the Audubon Nature Institute in New Orleans. Dresser’s goal, however, has not been to bring extinct animals back to life. Rather, she hopes to prevent animals from becoming extinct in the future. So far, she has produced clones of African wildcats by putting their DNA into the eggs of ordinary pet cats. The clones she has produced have not only been healthy, but have also given birth to kittens themselves. She believes that this kind of cloning will allow scientists to increase animal populations when it is necessary.

With this goal in mind, Dresser has been creating a “frozen zoo” of endangered animals. She has collected tiny pieces of skin containing the DNA of hundreds of different species from around the world. She keeps these frozen in small containers so that if it becomes necessary in the future, clones can be created. Not all experts agree with the idea of creating clones. Some of them say that it is more important to prevent damage to the environment. In this way, animals could continue to live where they are. Dresser agrees, but argues that if it is a choice between extinction and cloning, it is better to choose cloning.

(41) Why did some scientists criticize *Jurassic Park*?

- 1 They disliked using dinosaurs for entertainment.
- 2 They opposed the idea of creating fake dinosaurs.
- 3 They doubted that dinosaurs could ever be cloned.
- 4 They thought the dinosaurs in it were the wrong size.

(42) What is one way that scientists have been able to create animal clones?

- 1 By replacing the DNA in an animal's egg with the DNA of another animal.
- 2 By raising many different kinds of animals to be used for collecting DNA.
- 3 By gradually changing DNA to create a new animal.
- 4 By using electricity to create DNA in a laboratory.

(43) What is Betsy Dresser trying to do?

- 1 Increase the variety of pets that people keep.
- 2 Create new species that can avoid extinction.
- 3 Show that cloned animals are not healthy.
- 4 Make it possible to stop species from dying out.

(44) Why are some experts against cloning?

- 1 They are concerned about the effect of clones on other animals.
- 2 They worry that cloning will only be able to help a few species.
- 3 They think that it would be better to protect the places where animals live.
- 4 They believe that storing animal DNA for the future will be too expensive.

(45) Which of the following statements is true?

- 1 *Jurassic Park* seems much more realistic now than when it was first shown.
- 2 Dresser has collected clones of many different species from all over the world.
- 3 There is a possibility that even ordinary pets like cats will become endangered.
- 4 It is still impossible to create clones that are able to produce babies of their own.

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次の英文 **A**, **B** を読み、その文意にそって (26) から (33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び、その番号を解答用紙の所定欄にマークしなさい。

Slow Reading

Depending on our aim, there are various ways in which we can read books or articles. One method is known as “skimming.” We use this method if we want to find only the main ideas in a piece of writing. When we skim an article, we look through it quickly without paying attention to the (**26**). If we want to find some particular piece of information but are not interested in the overall meaning, then we can let our eyes pass over the text until we find the part we want. This is called “scanning.” However, if we want to understand fully what the author is saying, then we need to read the words slowly and carefully. This is known as “in-depth” reading, and according to a number of studies, it is becoming increasingly uncommon today.

Many people blame the Internet for the decline in in-depth reading. Nicholas Carr, a well-known writer on technology, argues that there are three features of the Internet that (**27**) in-depth reading. The first is the fact that people often use links to jump from one website to another on the Internet. The second is that people often interrupt their reading to check their e-mail or the news. The third feature is that much of what we read on the Internet is very short anyway. The result is that it has become (**28**) for people to concentrate on longer pieces of writing.

A number of writers and scholars have become so concerned about this situation that they are arguing we need to start a “slow reading” movement. This idea is based on the “slow food” movement, in which people take time to prepare and eat food together. None of the supporters of “slow reading” think we can (**29**) the Internet completely. Instead, they recommend that we turn off the computer once or twice a week, settle down with a book, and read with concentration for a few hours.

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|------|---------------------|------------------|-------------------|-------------------------|
| (26) | 1 tasks | 2 details | 3 pictures | 4 advertisements |
| (27) | 1 include | 2 improve | 3 mention | 4 discourage |
| (28) | 1 harder | 2 luckier | 3 fairer | 4 safer |
| (29) | 1 write down | 2 try out | 3 give up | 4 show off |

Hidden Treasures

Over time, many ships have sunk to the bottom of the ocean. Some of these sank because of bad weather, and others because they were attacked by enemies. The remains of these ships are known as shipwrecks, and the United Nations has estimated that there may be as many as 3 million of them around the world. Many shipwrecks are (**30**). Sometimes there is gold and other kinds of treasure inside them. But even when what they contain is less precious, they can still provide important information about how people lived in the past.

Recently, there have been many arguments about how shipwrecks should be treated. One reason is that new technology is making it much easier to (**31**) them. Underwater robots, in particular, are allowing people to search in deeper parts of the ocean than ever before. Now, a number of commercial companies are using such technology to find more and more shipwrecks. Under international law, these companies can keep what they find on the sunken ships except when the ships are ones that were owned by a government.

Many historians, however, are opposed to such (**32**) searches. They say that shipwrecks should not be disturbed by treasure hunters but protected by governments so that scholars can study them just as they are. This is partly because treasure hunters sometimes damage objects they find. But it is also because many objects are preserved better by being underwater. (**33**), commercial companies argue that modern fishing techniques and pollution mean that many shipwrecks are at risk underwater. They say they are actually helping historians by finding shipwrecks and saving the objects in them from damage. Nevertheless, in 2001, the United Nations introduced an international agreement stating that all shipwrecks should be left in the oceans. By 2010, over 30 countries had signed the agreement.

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|------|---------------------|-----------|----------------|---------------|
| (30) | 1 crowded | 2 secret | 3 convenient | 4 valuable |
| (31) | 1 locate | 2 repair | 3 design | 4 donate |
| (32) | 1 gentle | 2 private | 3 confusing | 4 traditional |
| (33) | 1 As a result | | 2 Furthermore | |
| | 3 On the other hand | | 4 For instance | |

Lessons from the Ozone Hole

In May 1985, scientists from the British Antarctic Survey shocked the world by announcing the discovery of a huge hole in the ozone layer above the Antarctic. Ozone is a gas usually formed from oxygen that is high in the earth's atmosphere. The ozone layer is essential to life on our planet because it protects us from harmful ultraviolet rays produced by the sun. Without the ozone layer, these rays would cause cancer and other diseases in both human beings and animals.

The world's reaction to the scientists' discovery was surprisingly quick. Data showed that the ozone hole had been created by gases called chlorofluorocarbons (CFCs), which were widely used in machines such as air conditioners and refrigerators. By 1988, 46 governments had signed the Montreal Protocol, a treaty that banned the use of these gases. Eventually, this treaty was signed by every member country of the United Nations—the only treaty ever to be signed by so many nations. It will be a long time before the ozone hole disappears completely, but it soon stopped growing. Persuading the international community to adopt the Montreal Protocol was considered to be a great achievement for the environmental movement.

According to scientists, there were two reasons why people responded so quickly to the danger. The first was that it was easy to understand. If the hole kept growing, it would threaten life everywhere. The second reason was that it was easy to replace the CFCs with hydrochlorofluorocarbons (HCFCs), gases that do much less damage to the ozone layer. Unfortunately, though, HCFCs have their own problems. Not only do they contribute to global warming, but they may also be a major factor in the development of acid rain.

Does this mean that the ban on CFCs was a bad idea? The ozone hole was a real danger and, without the ban, it would have quickly grown to cover the whole planet. Stopping its growth was a major victory. Nevertheless, the difficulties caused by HCFCs show how complicated environmental problems can be. Above all, they demonstrate how important it is for us to find a way to control the main cause of global warming—the harmful gases that we keep putting into the atmosphere.

(37) The ozone layer is important because it

- 1** helps preserve the oxygen necessary for life.
 - 2** protects us from rays that can cause serious diseases.
 - 3** allows us to calculate the amount of pollution in the air.
 - 4** stops the earth's atmosphere from being damaged by the sun.
-

(38) Why was the Montreal Protocol seen as a great success for the environmental movement?

- 1** It forced governments to do more research on the ozone layer.
 - 2** It provided data showing that CFCs created the ozone hole.
 - 3** It attracted attention to the damage being done to nature.
 - 4** It was agreed to by more countries than any other treaty.
-

(39) What was one reason CFCs could be banned so quickly?

- 1** Scientists were becoming concerned about acid rain.
 - 2** People recognized that global warming was a real problem.
 - 3** A gas that was not as harmful to the ozone layer was available.
 - 4** They were being produced in only a few places around the world.
-

(40) The ban on CFCs was a good idea because

- 1** it led to a reduction in global warming.
- 2** it saved the planet from a serious threat.
- 3** it reduced the amount of HCFCs in the air.
- 4** it showed how complicated nature is.

Natural Solutions

Malaria is a serious disease that affects millions of people every year. Malaria is spread by female mosquitoes, and one obvious way to fight it is to reduce the number of mosquitoes. This can be done very effectively by using chemicals that kill them. In fact, since the 1950s, this method has resulted in a large reduction in the number of malaria cases. The chemicals used to kill mosquitoes, however, have various disadvantages. Not only are they expensive, but they are often bad for the environment. Moreover, over time, mosquitoes gradually stop being affected by them.

For these reasons, scientists have recently been looking at alternative methods of controlling mosquitoes. One of these is using fish. Mosquitoes lay their eggs in water, and the eggs then turn into tiny worms that live in the water for one or two weeks. Some kinds of fish eat these worms, so introducing these fish into lakes and ponds can lead to fewer mosquitoes. Projects carried out in India have found that, depending on the kind of fish, this method can reduce the number of mosquitoes by over 90 percent.

Using fish to control mosquitoes has many advantages. One of these is that it does no damage to the environment. Another is that the fish reproduce by themselves, making this a very cheap method of fighting malaria. In addition, some of these fish can be caught and sold, meaning that local people can actually earn an income from them. All of these factors are especially important in the developing countries where malaria is still common.

Using nature to control nature in this way is known as “biocontrol.” Biocontrol itself is not a new idea, but scientists are now doing more research on it. One problem with using fish has been that it must be limited to permanent bodies of water, such as lakes. Mosquitoes, though, often lay their eggs in pools of rainwater that later dry up. Scientists have now found a kind of fish in the African country of Tanzania that can survive even when these pools are dry. When the rain comes, the fish eat the mosquitoes. Many experts believe that this kind of research is giving biocontrol a bright future as a way to fight diseases.

(41) How has the number of malaria cases been reduced since the 1950s?

- 1** By treating people with new medicines.
- 2** By destroying the places where mosquitoes live.
- 3** By getting rid of mosquitoes with chemicals.
- 4** By stopping mosquitoes from laying eggs.

(42) Why are scientists introducing certain fish into lakes and ponds?

- 1** They eat the worms that turn into mosquitoes.
- 2** They produce eggs that help to kill the mosquitoes.
- 3** They reduce the number of worms that mosquitoes eat.
- 4** They make it easier to see where mosquitoes are hiding.

(43) What is one advantage of using fish to fight malaria?

- 1** They remove chemicals that pollute the water.
- 2** They can be used by farmers to grow more crops.
- 3** They are not affected by the disease themselves.
- 4** They can help local people to make money.

(44) Why are the Tanzanian fish especially good for biocontrol?

- 1** They can survive without a permanent body of water.
- 2** They are already known to most people around the world.
- 3** They cost less to transport than other kinds of fish.
- 4** They live longer in lakes than most other kinds of fish.

(45) Why do many experts think that biocontrol has a bright future?

- 1** Research is explaining more clearly the causes of diseases.
- 2** The number of fish has increased in African lakes and ponds.
- 3** Scientists are finding new ways to use nature to control diseases.
- 4** Natural chemicals are often more effective than man-made ones.

3

A

次の英文 **A**, **B** を読み、その文意にそって (26) から (33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び、その番号を解答用紙の所定欄にマークしなさい。

Using Energy Wisely

Over the last few decades, the negative effects on the environment of burning fossil fuels have become increasingly obvious. As a result, governments have been trying hard to get people to reduce the amount of energy they use. A recent survey carried out in the United States, however, has shown that many people are (**26**) about the best way to do this.

When people were asked what would be the best way to cut down on energy use, the most common answer was to turn off lights as much as possible. According to experts, however, this saves very little electricity. On the other hand, few people suggested changing to a car that uses less fuel. This, though, would make a much greater (**27**) to the environment. Another effective way to save energy is to put material inside the walls of a house so as to keep heat from escaping. This answer, too, was given by very few people.

Why do so many people have the wrong idea about what they should do to protect the environment? According to Shahzeen Attari of Columbia University in New York City, one of the researchers who carried out the survey, there are various reasons. One is that experts have not been good at informing the public about what they should do. Another is that most people think simply of continuing to use the things they usually use, but less often. In fact, if people were to change to newer (**28**), they could carry on living in the same way but still use much less energy. For example, if people change to new kinds of highly efficient light bulbs, they can leave them on all the time and still use much less electricity than before. According to Attari, it is important that governments (**29**) people to invest in the new fuel-saving inventions that are now available.

- | | | | | |
|------|---------------------|--------------------|---------------------------|-----------------------|
| (26) | 1 honest | 2 mistaken | 3 excited | 4 disappointed |
| (27) | 1 complaint | 2 challenge | 3 payment | 4 difference |
| (28) | 1 technology | | 2 responsibilities | |
| | 3 locations | | 4 communication | |
| (29) | 1 reserve | 2 delay | 3 replace | 4 encourage |

Healthy Messages

Cell phones have become an important part of most people's lives. According to one survey, 86 percent of all American adults today have a cell phone, and 40 percent use their cell phones to connect to the Internet. Cell phones have had an impact on how people do business and also on how they spend their leisure time. (30), many medical professionals are now investigating how to use cell phones to help them look after people's health.

One quite simple way in which cell phones are being used in the medical field is to give instructions to patients about treatment. One clinic in Boston, for example, has begun sending text messages to patients whose skin is especially sensitive to the sun. The messages, which are based on weather forecasts, warn them when they must protect their skin with sunscreen lotion. According to the clinic, (31) use of sunscreen lotion has increased by 40 percent. Similarly, a hospital in New York City sends messages to teenage patients reminding them to take their medicine. If the hospital does not receive a reply, it then contacts the patient's parents and has them check that the medicine has been taken.

A more complicated experiment is being carried out by Dr. Richard Katz of Washington, D.C. His patients need to (32) the amount of sugar in their blood carefully. The patients measure their sugar levels with a special machine and then use cell phones connected to the Internet to send in the results each day. If the sugar level becomes too high, the patient receives a message explaining what to do to bring it down again. When the patient goes for a checkup, the doctor uses the daily measurements to help judge the patient's condition. According to Katz, the new system not only helps keep patients healthy but also saves (33). This is because it avoids the need for costly emergency treatment at the hospital.

- | | | | | |
|------|---------------|----------------|----------------|-----------------|
| (30) | 1 In addition | 2 For instance | 3 Nevertheless | 4 Unfortunately |
| (31) | 1 accidental | 2 wasteful | 3 correct | 4 secret |
| (32) | 1 hide | 2 watch | 3 taste | 4 absorb |
| (33) | 1 space | 2 power | 3 time | 4 money |

Born Athletes?

In recent years, scientists have learned more and more about the ways in which our DNA can influence our development and abilities. As a result, society is increasingly being faced with the problem of deciding how to use such research. Should we analyze people's DNA and use this information to make decisions about how they should live? A good example of this kind of problem has recently appeared in connection with sports.

In 2003, a group of scientists suggested that there could be a relationship between sporting ability and a gene known as ACTN3. Everyone has two ACTN3 genes, one from their mother and one from their father. These genes, however, come in two types. The R type of gene causes the body to build a protein that gives muscles a short burst of extra power. The X type stops this protein from being produced. Although many people have one of each type, the scientists suggested that people who are good at running quickly for a short time are more likely to have two R types. There is also some evidence that people who are good at running long distances, on the other hand, are more likely to have two X types.

Now, a company in the United States has started selling kits that allow parents to find out which kind of ACTN3 their children have. The parents take a small sample of their children's DNA and send it to the company for analysis. The company then sends back the results. The company says this service allows parents to find out which kinds of sports their children are likely to be better at. So far, a few hundred parents have bought the service.

Many experts, though, criticize this use of DNA research. Some say that the tests are useless because sporting ability involves many factors, not just one gene. Others are concerned that the tests may lead parents to force their children to play one sport rather than another. They say that the problem is not so much the DNA research itself. Rather, it is the fact that some parents seem to be determined to make their children into star athletes at any cost.

(37) What is one problem faced by society today?

- 1** Getting people to agree to DNA analysis.
 - 2** Preventing athletes from changing their DNA.
 - 3** Deciding the right way to use DNA research.
 - 4** Keeping the results of DNA tests private.
-

(38) How are the R type and X type of ACTN3 different?

- 1** The R type gives muscles more power for a short time.
 - 2** The R type is usually found together with another R type.
 - 3** The X type always comes from the DNA of one's mother.
 - 4** The X type makes it hard for people to do well at sports.
-

(39) Why do parents buy the ACTN3 kit?

- 1** To help their children find out about DNA analysis.
 - 2** To decide what school to send their children to.
 - 3** To learn about their children's ability at sports.
 - 4** To find out about their children's likes and dislikes.
-

(40) Some experts say that

- 1** DNA tests should be carried out by doctors rather than parents.
- 2** ACTN3 is just one of many things that affect sporting ability.
- 3** children should spend much less time on sports than they do now.
- 4** the testing kits give some children too much of an advantage.

Preserving Plants

Every year, over a million people visit Kew Gardens, a large park in London that originally belonged to Britain's Royal Family. Visitors go there to enjoy the park's beautiful trees and flowers. What many visitors do not realize, however, is that Kew Gardens is also home to the world's largest collection of plant samples. These are kept in a special building known as the Herbarium. Today, this collection is playing an important role in saving wild plants from extinction.

The Herbarium was founded in 1852. The first director of Kew Gardens, William Hooker, determined to collect examples of plants from all parts of the globe. He sent out collectors to gather samples for the Herbarium. These samples were then dried and glued to large sheets of paper. On each sheet, a note was made of where and when the plant had been found. This work continues today, and the Herbarium now employs 180 scientists, who gather up to 50,000 new samples every year.

Recently, scientists at Kew Gardens decided to carry out a survey to discover how many plants are threatened by extinction. The world contains a huge number of plant species, and it would not have been practical to check on all of them. Instead, the scientists decided to create five lists that would be representative of the world's five major groups of plants. Using the Herbarium's collection, the scientists chose 1,500 species at random from each group. They then asked experts around the world to find out whether these species were in danger. Based on the answers, they were able to get a better idea of the threat to the plants in each group.

The survey has shown that plants today are even more under threat than animals. This is something that many scientists had already suspected to be true. In fact, the survey suggests that around one-fifth of all plant species are in danger. The main cause is the destruction of their habitats by humans. According to the present director of Kew Gardens, Stephen Hopper, the importance of plants has been overlooked for far too long. He hopes that the list will allow conservationists to decide which plants and habitats need the most protection, helping to preserve the great variety of plants in nature.

(41) The Herbarium at Kew Gardens

- 1** will soon have to close unless more people start visiting it.
- 2** is used to provide the park's beautiful trees and flowers.
- 3** was built for plants that belong to the British Royal Family.
- 4** contains the biggest collection of plant samples in the world.

(42) After William Hooker became director of Kew Gardens, he

- 1** gave plant samples from Kew Gardens to experts in other countries.
- 2** sent people to different places around the world to gather plant samples.
- 3** employed scientists to find new ways to display dried flowers.
- 4** built the Herbarium to keep plants that were in danger of extinction.

(43) The scientists at Kew Gardens

- 1** tried to find out how many endangered plants there are in the world.
- 2** recorded detailed information about 1,500 different extinct plants.
- 3** discovered a large number of plants that had never been seen before.
- 4** based their research on the opinions of visitors to the Herbarium.

(44) What does Stephen Hopper think?

- 1** Many plant habitats are being taken over by endangered animals.
- 2** Kew Gardens should give more space to plants that are in danger.
- 3** People have not paid enough attention to protecting plant species.
- 4** Scientists often overlook the need to search for new plants.

(45) Which of the following statements is true?

- 1** People who answered the survey made many suggestions on how to conserve plants.
- 2** Stephen Hopper was asked to try to get the dried plants in the Herbarium to grow again.
- 3** Plants that were once thought to be extinct have been found growing at Kew Gardens.
- 4** The fact that plants were more endangered than animals was not a surprise to many scientists.

3

A

次の英文 **A**, **B** を読み, その文意にそって (26) から (33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

The Ikotoilet

People in developed countries have become used to the presence of clean toilets and bathrooms. In the developing world, however, the situation is very different. A good example is Kenya, where more than half of the population (**26**) access to toilets and clean water. One reason for this is the rapid growth of urban areas. As more and more poor people have moved from the countryside to cities, huge slums have developed. Homes in these very poor areas do not usually have any water or toilets.

David Kuria, a Kenyan architect, has come up with one solution to this problem: the Ikotoilet. It is a building that contains toilets, showers, and drinking fountains, all of which can be used for a very small amount of money. Besides being (**27**), the Ikotoilet has two other important features. The first is that it is designed to do minimum damage to the environment. The toilets use as little water as possible, and the waste produced is recycled as fertilizer. The second is that the Ikotoilets are designed to be fun, with music, a small shop selling refreshments, and other facilities. Kuria hopes that these things will make the Ikotoilets (**28**) places to visit.

So far, Ecotact, Kuria's company, has built 45 Ikotoilets around Kenya. Each one costs about \$20,000 to build and \$3,000 a year to run. The Ikotoilets not only provide a popular service but also give jobs to a total of 260 people. The real gain for Kenya, however, is much bigger. As in other developing countries, the shortage of clean water and toilets is one of the main causes of disease. Because of such disease, many children miss school, and many workers have to take time off. Preventing disease, (**29**), is important for a strong economy. Kuria is now planning to start building his Ikotoilets in many more African countries.

- | | | | | |
|------|--------------------|----------------------|-------------------|----------------------|
| (26) | 1 gives | 2 rejects | 3 lacks | 4 controls |
| (27) | 1 obvious | 2 traditional | 3 private | 4 inexpensive |
| (28) | 1 enjoyable | 2 temporary | 3 separate | 4 official |
| (29) | 1 by chance | 2 for example | 3 though | 4 therefore |

Holes in the Clouds

Have you ever had your flight delayed at an airport because of bad weather? If so, there may be a simple explanation. A recent study by a team of scientists has shown that at least some of the rain and snow that falls at airports is actually caused by airplanes taking off or landing. (**30**), the way this happens also explains the existence of strangely shaped holes in clouds that have puzzled scientists for decades.

The leader of the team, Andrew Heymsfield, got the idea for his research when he visited Denver International Airport, in the United States. While he was there, he noticed a narrow snowstorm that followed exactly the path of two airplanes that had taken off. He became sure that the airplanes had made the snow fall. In order to (**31**) this, the scientists in the team studied the effect of airplanes on clouds using satellite photographs, weather forecasts, and airport data. They discovered that when an airplane flies through a certain type of very cold cloud, it causes the temperature of the air in the cloud to drop further. This leads the water in the cloud to freeze and fall to the ground as snow or rain.

After the airplane has flown through the cloud, more and more of the cloud turns into rain or snow. This process continues for up to 90 minutes and leaves behind a large hole. Many scientists had noticed these holes in clouds before and wondered what made them. Now, we know the (**32**). According to Heymsfield, airplanes only affect clouds in this way under certain conditions. For this reason, the phenomenon probably does not lead to a dramatic increase in the amount of rain or snow that falls worldwide. Still, it has the result that airports are (**33**) than they would be otherwise.

- | | | |
|------|-----------------|-----------------|
| (30) | 1 What is more | 2 Unfortunately |
| | 3 After a while | 4 Instead |
| (31) | 1 reverse | 2 prove |
| | 3 achieve | 4 avoid |
| (32) | 1 opposite | 2 difficulty |
| | 3 future | 4 answer |
| (33) | 1 closer | 2 wetter |
| | 3 noisier | 4 safer |

Smoother Sailing

In recent years, there has been huge growth in global trade. This, in turn, has led to a dramatic increase in the number of ships being used to carry goods. These ships consume an enormous amount of fuel. This not only costs a great deal of money but also results in a lot of carbon dioxide being released into the atmosphere. As a result, the shipping industry is searching hard for ways to reduce the amount of fuel it uses.

Since most of the fuel is used to move the ship through the water, one good way to reduce fuel costs is to make it easier for the ship to travel through the water. One solution is to use bubbles of air. Recently, some new ships have been built with an “air-bubble system.” In this new system, the bottom of the ship has small holes in it, and a pump pushes air through the holes. This creates bubbles that join together to form a layer. The layer of air helps the ship to go forward more smoothly, so it needs less fuel.

At first, one problem with this system was that the bubbles tended to escape easily. However, by creating thin barriers on both sides underneath the ship, it became possible to trap the air for longer. Although the pump used to push the air out also needs fuel, much more fuel is saved overall. With the barriers, the system can reduce the amount of fuel used by as much as 15 percent.

Despite these impressive savings, very few shipping companies adopted air-bubble systems until recently. This was because it costs a great deal of money to build a new ship, and companies try to use their existing ships for as long as possible. Now, a Dutch company called the DK Group has solved this problem. It has created an air-bubble kit that can be fitted to an existing ship in just two weeks. With growing pressure on the shipping industry to produce less carbon dioxide, it seems likely that many companies will invest in these kits.

(37) What is one problem facing the shipping industry?

- 1** The ocean has become increasingly crowded.
 - 2** It has to spend a lot of money on fuel.
 - 3** There are not enough goods being traded.
 - 4** Carbon dioxide has been affecting ships' engines.
-

(38) How does the air-bubble system work?

- 1** It helps water to flow through holes in the ship's bottom.
 - 2** It saves fuel by mixing it with water from the sea.
 - 3** It uses a pump to remove air from the ship's fuel.
 - 4** It creates a layer of air on the bottom of the ship.
-

(39) In the air-bubble system,

- 1** barriers under the ship help stop air bubbles from escaping.
 - 2** the fuel that is saved can be used to produce electricity.
 - 3** bubbles can keep the ship's bottom in good condition.
 - 4** the pump can prevent the ship from moving too quickly.
-

(40) Why did few companies start using air-bubble systems until recently?

- 1** The systems would make ships produce too much carbon dioxide.
- 2** The people who work on the ships need training to use the systems.
- 3** It was necessary to build new ships to use them.
- 4** It took too long to install the systems on existing ships.

Africa's Great Green Wall

The Sahara Desert, in Africa, is the world's largest hot desert, and it is getting bigger all the time. Researchers say that, partly because of global warming, the desert is now spreading southward by up to about 50 kilometers a year. This has made life very hard for people in the countries that are directly south of the Sahara. Now, however, a major plan to stop the desert's growth is about to be put into practice.

This plan is known as the Great Green Wall, and it involves the creation of a "wall" of trees 15 kilometers wide and almost 8,000 kilometers long. The Great Green Wall is intended to reduce damage from the sandstorms blowing off the Sahara and to help keep the soil stable and fertile. Its trees will provide local people with wood and other materials, and it will become a home for plants and animals. The trees will also help to remove carbon dioxide from the air. Eleven different countries across Africa have agreed to participate in the project.

The idea itself is not a new one. In fact, it was first suggested in the 1980s. The problem has been a shortage of money. This all changed in 2011, when a group of international organizations, including the Global Environment Facility (GEF), agreed to donate up to \$3 billion to the project. A number of NGOs representing local communities, however, are concerned about the plan's possible effects. They are especially worried about the idea of planting a large number of trees. They say this might mean introducing new kinds of trees from other areas that would damage local ecosystems and use up valuable farmland.

The GEF, however, says that these fears are unnecessary. They say that they are not just planning to have trees planted across Africa. Rather, they will require each country to consult with local citizens and come up with a plan that will improve people's lives without harming the environment. As the GEF points out, simply planting trees will not work unless local people have some reason to look after them. By choosing trees that offer a source of income, such as fruit trees, the GEF believes that the Great Green Wall will continue to help people long after it has been completed.

(41) What problem are some African countries having?

- 1** Plans to improve the environment are not working.
- 2** Their land is being taken over by the Sahara Desert.
- 3** Researchers cannot get enough money to study there.
- 4** Many people have to move because of the increased heat.

(42) The Great Green Wall will

- 1** make it harder to travel between some countries.
- 2** benefit plants and animals as well as humans.
- 3** require a lot of wood and other materials to build.
- 4** reduce the number of sandstorms inside the Sahara Desert.

(43) Why did it take a long time to start the Great Green Wall project?

- 1** Local communities did not want any more trees.
- 2** Nobody could find trees that were suitable for the wall.
- 3** There was not enough money available to plant the trees.
- 4** Some people were worried that the trees would create carbon dioxide.

(44) What is one reason the GEF thinks the Great Green Wall will be a success?

- 1** The same kinds of trees can be planted in every country.
- 2** The trees will help people living nearby to make more money.
- 3** The organization will be responsible for looking after the trees.
- 4** The plan has been approved by NGOs representing local people.

(45) Which of the following statements is true?

- 1** Some of the trees that the GEF planted in the 1980s are still growing in Africa.
- 2** The purpose of the Great Green Wall is to provide wood for new buildings.
- 3** Some people worry that trees from outside the local area will cause problems.
- 4** The Great Green Wall project was started by NGOs representing local communities.

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次の英文 **A**, **B** を読み、その文意にそって (26) から (33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び、その番号を解答用紙の所定欄にマークしなさい。

Image Problem

The small fish known as piranhas that live in South America generally have a very bad reputation. People think of them as aggressive and dangerous, and they are often shown attacking and eating people in novels and films. Historians say this (**26**) image began when Theodore Roosevelt, a president of the United States, visited Brazil in 1913. While he was there, local fishermen warned him not to go into a river because he might be attacked by piranhas. Roosevelt refused to believe their story. The fishermen then pushed an old cow into the river, and hundreds of piranhas attacked and ate it in a few minutes. Later, Roosevelt wrote about this incident in a book.

According to researchers who have studied piranhas, though, this frightening reputation is exaggerated. In fact, piranhas hardly ever attack people, and no case is known of them having killed a human being. They (**27**) eat only the bodies of fish and animals that have died, or they take small bites out of other fish. They will sometimes form a large group and attack another animal, but this happens only when they are extremely hungry.

What about the dramatic event that Roosevelt witnessed? Research has shown that this was actually (**28**) by Brazilians to make the piranhas seem dangerous. Weeks before Roosevelt's visit, the fishermen collected piranhas and kept them without any food in one part of the river. When they put the old cow in the river, the fish attacked it because they were starving. (**29**), there have been more reports of piranhas biting people in recent years. Experts believe this is because the piranhas have become used to fishermen throwing unwanted parts of fish into the rivers. When someone puts in a hand or foot, piranhas sometimes mistake this for the food they are waiting for.

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|------|----------------|------------------|-----------------|-------------|
| (26) | 1 scientific | 2 embarrassing | 3 scary | 4 temporary |
| (27) | 1 briefly | 2 secretly | 3 unfortunately | 4 usually |
| (28) | 1 organized | 2 copied | 3 recorded | 4 shortened |
| (29) | 1 As a result | 2 In other words | | |
| | 3 For instance | 4 Nevertheless | | |

London's Green Bridge

One of the most popular alternatives to using fossil fuels is solar energy. As concern about global warming grows and the cost of solar panels declines, more and more companies are starting to use solar energy for their buildings and factories. Recently, work has begun in London on converting an old train station into a new “green” building. The new station will get about half the energy it needs from solar panels. Many environmentalists have (**30**) the project. They say that it is a good example of an approach to energy suitable for the 21st century.

The station will use an old bridge—Blackfriars Railway Bridge, which crosses the River Thames—as part of its design. Platforms for passengers will be built along both sides of the bridge. A long roof will be built over the bridge with 4,400 solar panels on it, enough to produce about 900,000 kilowatt-hours of electricity a year. The use of solar energy will reduce the amount of carbon dioxide released into the air and also save on electricity costs. The station will be (**31**) for the environment in other ways, too. There will be a system to collect rainwater for use in the station, and special devices will reflect natural sunlight into the station.

The new design, however, raises one difficult issue. Blackfriars Railway Bridge was built about 140 years ago and has been (**32**) by the British government as a historic building. The new station, however, will completely change the way the bridge looks. This would not normally be allowed, but the government has given permission for the new project. Many people, though, are worried about placing modern technology, such as solar panels, on historic buildings. They say that there are other ways to save energy without changing the (**33**) of old buildings and other structures. The company responsible for building the new station, however, insists that today saving energy must be our first priority.

- | | | |
|------|-------------|-----------------|
| (30) | 1 replaced | 2 misunderstood |
| | 3 ignored | 4 welcomed |
| (31) | 1 basic | 2 helpful |
| | 3 risky | 4 complicated |
| (32) | 1 improved | 2 exported |
| | 3 protected | 4 offered |
| (33) | 1 position | 2 appearance |
| | 3 security | 4 benefit |

A New Trend in Volunteering

Voluntary Service Overseas (VSO) is the name of an organization that was founded in the United Kingdom in 1958. The original aim of VSO was to send British 18-year-olds abroad so that they could learn about the problems of people in poorer parts of the world. Over time, however, VSO's focus changed, and it began sending mostly older people with professional skills. Although these volunteers are not paid, VSO covers all their expenses, including travel. The organization gradually grew so large that it founded partner agencies in countries around the world.

Unlike some volunteer agencies, VSO accepts volunteers no matter what country they are from. This fact, combined with the existence of partner agencies in a number of developing countries, has given rise to a new phenomenon. When we think of volunteers, we normally imagine people from wealthy countries helping people in poorer countries. Increasingly, though, VSO is recruiting volunteers from one developing country and sending them to help in another. A doctor in India, for example, might be sent as a volunteer to look after sick people in an African country.

The most dramatic example of this is probably in the Philippines. Volunteers from developed countries regularly come to the Philippines to help with development projects. But since VSO Bahaginan, the Philippines' VSO partner organization, was founded in 1999, it has sent over 600 Filipino citizens to other developing countries to work on projects. Most of these volunteers are professionals between the ages of 35 and 45.

Naturally, many people find it strange that these volunteers go abroad instead of working to help people in their own country. According to Rachel Nalus Quintos, VSO Bahaginan's chief communications officer, sending Filipino volunteers abroad has two advantages. One is that they gain international experience that can then help them in their own careers. The other is that they often come back to the Philippines after their time abroad determined to do more to end poverty in their home country. In this sense, volunteering abroad has the effect of turning them into true global citizens.

(37) What is one thing we learn about VSO?

- 1** It grew larger by combining with other volunteer organizations.
- 2** It was created to give volunteers more professional experience.
- 3** It pays the transportation costs for volunteers to go to other countries.
- 4** It is the oldest volunteer organization in the United Kingdom.

(38) What is one thing that makes VSO different from some other volunteer organizations?

- 1** It does not form partnerships with other volunteer agencies.
- 2** It sends people of any nationality to help with projects.
- 3** It relies on money from both wealthy and developing countries.
- 4** It does not provide medical help to developing countries.

(39) VSO Bahaginan is an example of how

- 1** VSO volunteers today do not fit the pattern of VSO volunteers in the past.
- 2** a poorer country refused help from an international volunteer organization.
- 3** the Philippines receives more international volunteers than other areas in Asia.
- 4** many doctors are trying to focus more on public-health problems in Africa.

(40) What is one reason Rachel Nalus Quintos supports the activities of VSO Bahaginan?

- 1** It teaches people about the cultural traditions of the Philippines.
- 2** It helps the Philippines maintain peaceful relations with other countries.
- 3** It results in more efforts to help poor people in the Philippines.
- 4** It encourages more foreign volunteers to come to the Philippines.

A New Way to Wash

Until the invention of the modern washing machine, most people washed their clothes by hand. In the 20th century, however, washing machines became common, and washing clothes became much easier. Washing machines, though, have some serious disadvantages. One is that they need a great deal of clean water. Another is that, in many countries, hot water is used in the machines. This consumes a lot of energy. In the United Kingdom, for example, about one-third of household energy is used for heating water, including that used to wash clothes.

Now, a British company has invented a washing machine that solves these problems by using small plastic beads to absorb dirt and clean the clothes. First, a little water is added to the clothes to get them wet. Then, the beads are mixed with the clothes until they have taken away all the dirt and stains. When the process is over, the little beads are removed by the machine, and the clothes are left almost dry. Because the beads can be used over and over, the machine costs very little to operate.

The new machine is based on the research of Stephen Burkinshaw, a chemist at the University of Leeds in the U.K. For a long time, Burkinshaw has been studying ways to make plastic materials a different color. A few years ago, he noticed that many kinds of plastic, especially nylon, attract dirt from other objects. He found that although the nylon absorbs dirt in warm and wet conditions, the surface of the nylon remains clean. For this reason, beads made of nylon can be used repeatedly until they are completely full of dirt.

The new washing machine can save households a great deal of the money they spend on water and energy. At the same time, the machines are very good for the environment. The United States, for example, uses 1.25 billion metric tons of water on washing clothes every year. If the new machines were adopted there, 90 percent of this water could be saved. The saving of electricity, meanwhile, would help decrease the amount of carbon dioxide produced by power plants. For these reasons, the company is hoping that its new “waterless” washing machines will be widely used in the future.

(41) What is one problem with the washing machines used in many countries?

- 1 Many people do not have enough money to buy one.
- 2 They consume one-third of the water used in homes.
- 3 A lot of energy is required to heat the water they use.
- 4 They do not get clothes as clean as washing by hand.

(42) The small plastic beads used by the new washing machine

- 1 are separated from the clothes when the cleaning process has finished.
- 2 contain enough water and energy to wash the clothes.
- 3 stop the clothes from being damaged while they are being washed.
- 4 must be taken out to be cleaned each time the machine is used.

(43) What did Stephen Burkinshaw learn about nylon?

- 1 It absorbs dirt from other objects in certain conditions.
- 2 It causes other plastic objects to stick to wet clothes.
- 3 It can be made into clothes that can be washed very easily.
- 4 It changes color when it comes into contact with dirty clothes.

(44) What is one reason the new washing machines may be good for the environment?

- 1 People who use them would wash their clothes less often.
- 2 They could reduce the amount of carbon dioxide being released.
- 3 The plastic beads from them can be used for other purposes.
- 4 The water in them is recycled over and over until it is dirty.

(45) Which of the following statements is true?

- 1 Burkinshaw's research has already helped the United States to save a lot of water.
- 2 Burkinshaw's research was originally about ways to change the color of plastic materials.
- 3 The new machines can wash clothes more quickly than ordinary washing machines.
- 4 The new machines are better at washing nylon clothes than those made from other materials.

3

A

次の英文 **A**, **B** を読み, その文意にそって (26) から (33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Sailing into the Future

As global trade has increased, so has the importance of the shipping industry. Today, in fact, 87 percent of all goods traded internationally are carried by ships. Recently, though, the industry has come under pressure to change the way it operates. One reason for this is (26). Over the past few years, shipping companies have found it increasingly difficult to make a profit, mainly because of a sharp rise in fuel costs. Another reason is environmental. The shipping industry is responsible for a large amount of the world's carbon emissions. Governments are demanding that shippers cut emissions so as to lessen global warming.

It is not surprising, therefore, that some companies have begun to (27) at the sailing ships that carried goods many years ago. These ships used a form of energy that cost nothing—the wind—and they caused no carbon emissions. One such company is B9 Energy, which has designed a ship that uses sails about 60 percent of the time and a biofuel engine when there is not enough wind. The ship carries a much bigger cargo than a traditional sailing ship, and it is cheaper to run than a fossil-fuel ship.

(28) this, some experts question whether sailing ships will replace fossil-fuel ships. This is because the latter can carry many more goods. B9 Energy's sailing ship can carry 9,000 tons of cargo, but the biggest fossil-fuel ships can carry 100,000 tons. Thus, even if the cost of fossil fuels rises, ships that use such fuels will still remain a cheaper way to transport goods than sailing ships. David Surplus, the chairman of B9 Energy, (29). He points out that, one day, fossil fuels will run out. He thinks that when this happens, his company's ships will be in a strong position to replace the ships used today.

- | | | | | |
|------|--------------------|------------------------|---------------------|---------------------|
| (26) | 1 official | 2 moral | 3 economic | 4 doubtful |
| (27) | 1 sit down | 2 come straight | 3 ask around | 4 look again |
| (28) | 1 Before | 2 Despite | 3 Due to | 4 And with |
| (29) | 1 disagrees | 2 recovers | 3 delays | 4 returns |

A New Kind of Ownership

During the 20th century, developed countries experienced a “consumer revolution.” As living standards rose, ordinary people became able to buy their own homes and a range of consumer goods to put in them. At the same time, companies began to produce cheaper televisions, refrigerators, vacuum cleaners, and other household items. People bought these things not only because they made their lives easier and more enjoyable, but also because they were a sign of (**30**). Owning them showed that you were a successful member of society. Many people felt ashamed if they did not have such consumer goods in their homes.

Now, however, some experts in the United States say that a new attitude toward owning things is developing, especially among people in their 20s. They call this way of thinking “collaborative consumption.” This means that people are no longer trying to own as many things as possible. (**31**), they borrow them for short periods of time from other people. This new kind of consumption has been made possible by the Internet. For example, an Internet site called SnapGoods allows people to search for the owner of an item who will let them use it for a small amount of money or even for free. Rather than buying an expensive tool that you are unlikely to use again, you can go onto the site and look for someone living nearby who is willing to (**32**) you one.

More and more websites like SnapGoods are now being created, covering everything from cars to cameras. People who use such sites report many (**33**). An obvious one is financial. To avoid paying a large amount of money for something they will only use a few times, people can rent it cheaply just when they need it. Another benefit is that these websites allow people to meet others in their community. This system cuts down on waste and therefore is good for the environment.

- | | | | | |
|------|-----------|-----------|---------------|----------------|
| (30) | 1 trust | 2 status | 3 youth | 4 friendship |
| (31) | 1 Instead | 2 Overall | 3 By accident | 4 What is more |
| (32) | 1 find | 2 choose | 3 lend | 4 make |
| (33) | 1 ideas | 2 prices | 3 mistakes | 4 advantages |

A Strong Bond

When someone breaks a bone, a doctor can usually treat it so that it heals naturally. In some cases, though, the break is very complicated, and the different parts of the bone have to be held together with a metal pin or screws. In other cases, there are so many small pieces of broken bone that even this technique cannot be used. The ideal solution would be to create a glue that could be used on bones. So far, however, no glue has been invented that can work in the wet conditions of the human body.

One place where researchers have been looking for an answer is under the sea. There are various creatures, such as shellfish, that live in the sea and yet manage to stick to things. Since the 1980s, scientists have been studying these animals in the hope of finding a way to create a glue that would work inside the body. Unfortunately, no one has yet been able to make such a glue. Now, one scientist, Russell J. Stewart of the University of Utah in the United States, thinks that he may have found the answer.

Since 2004, Stewart has been studying the sandcastle worm, a small worm that lives in the sea. The worm gets its name from the fact that it builds a shelter out of sand and pieces of shell. The worm produces a glue from its head that it uses to stick the sand and shells together. This glue becomes hard when the amount of acid in the water around it changes. This is very different from normal glue, which hardens as it dries out in the air.

Stewart and his team of scientists have now also produced an artificial glue that hardens in water. Unlike the worm's glue, however, the glue that they have created hardens when the temperature rises. This would make it ideal for use inside the body, where the temperature is higher than in the surrounding air. The main problem is whether the body will accept the glue or not. Stewart has successfully tried it on rats, and he is confident that he will soon be able to produce a glue that works in humans.

(37) Doctors sometimes use metal pins when

- 1** it is too expensive to use glue to hold a bone together.
 - 2** some of the pieces of a bone are difficult to find.
 - 3** wet conditions make it impossible for a bone to fix itself.
 - 4** a bone is broken in such a way that it cannot heal naturally.
-

(38) Why are scientists studying sea creatures such as shellfish?

- 1** New bones can be grown inside of their bodies.
 - 2** Their shells can be used to treat people's injuries.
 - 3** Some of them are able to stick to things underwater.
 - 4** They produce stronger glues than humans can.
-

(39) The sandcastle worm

- 1** creates a kind of acid that protects it from enemies.
 - 2** is interesting to scientists because it has a very hard shell.
 - 3** takes air from the water's surface to make its home stronger.
 - 4** lives in a shelter that it makes from materials in the ocean.
-

(40) What is one reason why Russell J. Stewart thinks his glue may be useful?

- 1** It reduces the amount of acid in the ocean.
- 2** It can be produced in laboratories using rats.
- 3** It is not affected by the surrounding temperature.
- 4** It does not need air to become hard.

Naleds

Mongolia has an extreme climate. During the winter, temperatures drop to below -30°C , and in the summer, they rise to above 20°C . Like other parts of the world, though, Mongolia is being affected by global warming. In fact, the average temperature has been steadily rising for the last 60 years. Such warmer temperatures might seem to be a good thing, especially in winter, but in reality, they are having mainly negative effects.

Traditionally, most Mongolians have lived by keeping animals such as cows and sheep. These animals feed on the huge grasslands that cover most of the country. However, in the warmer months, the higher temperatures caused by global warming dry out the soil, making it hard for the grass to grow. As a result, there have been shortages of food for the cows and sheep, and more and more Mongolians have had to sell their animals and move into the cities.

Recently, a British-Mongolian group of companies has come up with an idea that they think will help lessen the effects of global warming. Robin Grayson, a geologist working for the group, has focused on what are called “naleds”—thick bodies of ice that form on rivers in the winter. After a river freezes over, pressure grows in the water below until the water breaks through the ice. This forms a new layer of ice, making the naled thicker. This happens again and again over the winter, with some naleds growing as thick as 7 meters.

Grayson’s idea is to create artificial naleds. Engineers will drill holes through the ice and pump up water from the river below. In this way, naleds can be created that are so thick they last the whole summer. These will then be used both to cool the air and to provide a steady supply of water. The group’s first project is to create a naled near the capital, Ulan Bator. They hope this will cool the city—leading to less energy use—and provide more water. Grayson thinks that, in the future, many other parts of the world with very cold winters are likely to adopt this new idea.

(41)

Mongolia's weather

- 1 has suffered little from global warming.
- 2 has been getting steadily cooler for many years.
- 3 is warmer than in many surrounding countries.
- 4 is very different in winter and in summer.

(42)

Why have many Mongolians moved into the cities?

- 1 The food is cheaper there than in the countryside.
- 2 There has not been enough grass to feed their animals.
- 3 They want to escape the cold weather of the grasslands.
- 4 Large groups of animals have damaged the countryside.

(43)

“Naleds”

- 1 were an early sign of global warming in Mongolia.
- 2 can be broken up into pieces and transported to other areas.
- 3 are created when water freezes repeatedly on top of a river.
- 4 become much thicker when they form in the summer.

(44)

What might be one good effect of Robin Grayson's idea?

- 1 It will stop rivers from freezing over during the winter months.
- 2 It will reduce the amount of energy needed to keep Ulan Bator cool.
- 3 It will help create products that Mongolia can sell to other countries.
- 4 It will provide a new way for people in colder countries to keep warm.

(45)

Which of the following statements is true?

- 1 Naleds do not melt completely during the summer if they are thick enough.
- 2 Mongolia's soil has not been greatly affected by the changes in the climate.
- 3 Grayson's project has already been tried out in a number of other countries.
- 4 Winters have gradually become much colder in Mongolia over the past few decades.

3

A

次の英文 **A**, **B** を読み, その文意にそって(26) から(33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Bringing Back the Fruit

Every spring, the trees along San Francisco's sidewalks are full of cherry, pear, and apple blossoms. These flowering trees, though, do not go on to produce any fruit. This is because they have been specially developed to grow only flowers. In fact, San Francisco does not allow trees that produce fruit to be grown along city sidewalks. However, not everyone (**26**) this policy. One group in particular, which calls itself the Guerrilla Grafters, is working to change these trees back into ones that will provide fruit.

The group is doing this by using a technique called grafting. Grafting means cutting off a small branch from one tree and attaching it to another tree so that the branch becomes part of it. The members of the Guerrilla Grafters have started taking branches from trees that produce fruit and grafting them onto the flowering trees on the sidewalks. They hope that this will eventually allow people in the city to get fruit for free. What they are doing, however, is (**27**). This means they have to graft the branches secretly.

Officials want the Guerrilla Grafters to obey the law for a number of reasons. They say that the fallen fruit can cause (**28**). This is because it makes the sidewalks slippery, and people fall and injure themselves. The fruit could also attract rats and other animals. Tara Hui, one of the Guerrilla Grafters, points out that the group asks people to take care of the trees that have been grafted. These people gather up any unwanted fruit. According to Hui, an important aim of their campaign is to change people's relationship with the city environment. Right now, most people do not (**29**) to public places. By taking care of these trees and benefiting from them, people are likely to develop a sense of responsibility for the parts of the city that are used by everybody.

- | | | | | |
|------|------------------------|-------------------------|-----------------------|------------------------|
| (26) | 1 improves | 2 discusses | 3 influences | 4 accepts |
| (27) | 1 generous | 2 illegal | 3 advanced | 4 realistic |
| (28) | 1 fights | 2 waste | 3 accidents | 4 hunger |
| (29) | 1 enjoy driving | 2 feel connected | 3 seem limited | 4 stop donating |

The Surui Project

The people of the Surui tribe live deep in the Brazilian rainforest. Like many other such groups, the tribe is struggling to maintain its traditional lifestyle in the modern world. Although a large area of the rainforest officially belongs to them, there are many (**30**) to their tribal home. Some companies cut down the Surui people's trees for wood even though it is against the law. Others dig mines to look for gold. A few years ago, however, the chief of the tribe, Almir Narayamoga Surui, came up with a surprising new way to keep the tribe's land (**31**). His idea was to use information technology.

The chief decided to use Google Earth. This is a program that uses photographs taken by satellites along with information from other sources to create a map of the earth. By downloading the program, individuals can explore the earth using a computer or smartphone. The chief visited the headquarters of Google in the United States and asked the leaders of the company for their cooperation. The company agreed to provide detailed satellite photographs of Surui lands. Google also gave the Surui people smartphones that allowed them to take photographs of anybody cutting down trees. The Surui hope that such (**32**) will force the Brazilian government to take action against people damaging the forest.

The Surui also used the smartphones to create a detailed "cultural map" of their lands. This is a map that consists of photographs, videos, and 3-D models of important traditional sites. Google has agreed to include the map, which was completed in 2012, in their program. The Surui people hope that the map will make more people interested in supporting them in their struggle. (**33**), Google believes that the Surui Project will provide a helpful model for other groups around the world struggling to protect their lands.

- | | | | | |
|-------------|----------------|--------------------|-------------|------------|
| (30) | 1 threats | 2 invitations | 3 additions | 4 keys |
| (31) | 1 hidden | 2 cheap | 3 safe | 4 useful |
| (32) | 1 questions | 2 essays | 3 quality | 4 evidence |
| (33) | 1 For instance | 2 In contrast | | |
| | 3 Nevertheless | 4 At the same time | | |

Learning from One's Mistakes

When Bette Nesmith Graham dropped out of high school in the United States, few people would have guessed that one day she would become a millionaire. Graham got a job as a secretary and took classes at night to finish high school. She was good at her job and eventually became a secretary at a bank. At this time, in the 1950s, electric typewriters had just been invented, and secretaries were finding it difficult to get used to them. Because it was almost impossible to erase mistakes, Graham and her co-workers often had to retype a whole page just because of one mistake.

One day, Graham was watching some painters who were decorating the bank's windows for Christmas. She noticed that when they made a mistake, they painted over it instead of going to the trouble of wiping it off. This gave her an idea. When she went home, she mixed together some paints until she had created a liquid that was the same color as the office's paper. At work, she used this liquid to paint over her mistakes and typed over them when it dried.

Soon, other secretaries began asking Graham for some of the liquid. She filled more bottles and gave them out. She called her invention Mistake Out, and after the liquid became more popular, she decided to start a business from her home. She worked hard to make her product better and began selling it. Eventually, the bank found out about her business, and she lost her job. However, this gave her more time to work on building up her company.

In 1956, she renamed the product Liquid Paper. By 1968, she was selling a million bottles of it a year. Although Graham was very successful, she was not only interested in making money. She also wanted to help other women in the workplace, so she set up a library and a childcare center in her factory. She also used some of her fortune to create two organizations to help women find new ways to get jobs. Although typewriters have now been replaced by computers, Liquid Paper still remains popular as a convenient way to correct mistakes in handwritten documents.

(37) What was one problem Bette Nesmith Graham had in the 1950s?

- 1** People did not think she was a hard worker.
 - 2** Typewriters were still very expensive to buy.
 - 3** She spent a lot of time typing pages over again.
 - 4** She was too busy to get a high school degree.
-

(38) The painters at the bank

- 1** lent Graham some paints for her invention.
 - 2** washed the bank's windows with a new cleaning liquid.
 - 3** taught Graham a way of making Christmas decorations.
 - 4** made their job easier by covering over their mistakes.
-

(39) Why did Graham lose her job at the bank?

- 1** Her boss was angry because she made a lot of mistakes.
 - 2** Her own company took away many of the bank's customers.
 - 3** The other secretaries became jealous of her success.
 - 4** The bank discovered that she had started her own business.
-

(40) What is one way Graham used her money to help people?

- 1** She offered childcare services to the women who worked for her.
- 2** She gave free bottles of Liquid Paper to her factory workers.
- 3** She paid for research that made personal computers better.
- 4** She hired workers who had lost their jobs at other companies.

Healthier Citizens

In many countries, governments are trying to get citizens to eat more healthily. One way in which governments are trying to do this is by increasing taxes on foods that are unhealthy. For example, Denmark, a country in Europe, taxes products that contain a lot of sugar, such as ice cream and chocolate. In 2011, the Danish government went one step further and introduced a tax on the fats contained in foods like butter and cheese. However, experts do not agree whether such taxes work.

Some health experts have always disagreed with taxing fats. They say we need to consider each food separately and not simply worry about how much fat it contains. For the same reason, Danish dairy farmers felt that the government was wrong to say that the foods they produce are unhealthy. Cheese, for example, contains a lot of fat, but it does not seem to lead to health problems in the way other kinds of fat do. Such critics believe that the government should instead try to make healthy foods, such as vegetables, cheaper.

There have also been economic problems because of the tax. The Danish government thought that if the cost of fatty foods went up, people would not buy as much of them. However, some foods such as butter and cheese are an important part of the traditional Danish diet. Since these could not be bought in Denmark at reasonable prices, many Danish people began shopping in nearby countries. One study discovered that as many as 48 percent of Danish people were going to Germany or Sweden to do shopping, and the Danish economy lost about \$1.8 billion because of this.

For these reasons, the Danish government decided to get rid of the tax after only one year. However, many doctors say that one year is not long enough to know if the tax helped Danish citizens' health. They claim that the Danish government is worried more about the economy than it is about the Danish people. But even though the tax did not work in Denmark, some countries have not given up on the idea. The United Kingdom and France are thinking of trying similar taxes, and Hungary has already put a tax on fatty foods.

(41)

What is one way governments are trying to make people healthier?

- 1 By promoting low-fat versions of many popular foods.
- 2 By forcing them to pay more for certain kinds of food.
- 3 By decreasing the amount of sugar allowed in products.
- 4 By creating laws concerning how much people are allowed to eat.

(42)

Dairy farmers in Denmark

- 1 felt that they were being made to produce too many products.
- 2 were worried that the government was planning to ban cheese.
- 3 argued that their products are healthier than the government said.
- 4 admitted that their products contained an unhealthy amount of fat.

(43)

What was one result of taxing foods that have a lot of fat?

- 1 Many people avoided paying the tax by shopping abroad.
- 2 Many people began eating traditional Danish food.
- 3 It made healthy foods like fruit and vegetables cost more.
- 4 It caused people in Germany and Sweden to spend less money.

(44)

Many doctors think that the Danish government

- 1 could have ended the tax on fats much sooner.
- 2 is more concerned about money than the health of its citizens.
- 3 has enough information to see if taxes on unhealthy foods are effective.
- 4 should ask other countries to make laws against selling fatty foods.

(45)

Which of the following statements is true?

- 1 Some people think that making healthy foods cheaper would work better than a tax.
- 2 Health experts say that we should eat more dairy products than other foods.
- 3 Some kinds of fruit and vegetables are as unhealthy as meat.
- 4 Many people buy unhealthy foods to avoid paying high taxes.

3

A

次の英文 **A**, **B** を読み, その文意にそって(26) から(33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Standing Up for Health

The development of computer technology has made our lives more convenient in many ways. This is especially true at the office. Nowadays, with a computer, people can create documents, look up facts, send messages, and even hold meetings—all without leaving their desks. This (**26**) efficiency, though, also has a dark side. Many people now sit down all day in the office. While this may seem comfortable at first, it often leads to neck or back pain, and it can cause people to develop heart disease and other serious problems.

One solution that is becoming popular is tall desks that can be used while standing. Standing can result in weight loss, a stronger back, and even better concentration. (**27**), such standing desks are not new. In the past, many famous writers and politicians used these. Moreover, some schools even used to have standing desks for students. For a while, these desks were becoming rare, but they are now becoming popular again. Many furniture companies say that sales of standing desks have gone up greatly.

Not everybody, however, thinks that standing desks are the (**28**). Standing all day in one place is also not very good for the health. Additionally, many people find standing desks difficult to get used to. In offices which have both standing and sitting desks, most people actually choose to sit for most of the time. According to some experts, a better idea is for us to change our working (**29**). Instead of sending an e-mail to someone in the same office, we should go and talk to them. We should also hold meetings while standing up or moving around. In this way, people can get more exercise even when they are working.

- | | | | | |
|------|-------------------------|--------------------------|-----------------|--------------------|
| (26) | 1 fight against | 2 increase in | | |
| | 3 explanation of | 4 complaint about | | |
| (27) | 1 Therefore | 2 Otherwise | 3 If so | 4 In fact |
| (28) | 1 excuse | 2 answer | 3 reward | 4 reason |
| (29) | 1 habits | 2 spaces | 3 hours | 4 contracts |

A Growing Problem

Global warming is creating problems not only for human beings but also for many plants and animals. Most people know that climate change is a huge threat to the future of many land animals, but what about fish? According to new research by scientists at the University of British Columbia, climate change will have some (**30**) impacts on fish, too. The scientists are saying both that the average size of fish in the oceans will get much smaller and that their numbers will greatly decrease by the year 2050.

As the oceans heat up, the size of fish will be changed in two ways. First of all, the amount of oxygen in the water will go down, which will keep the fish from growing to normal size. Secondly, the average body temperatures of the fish will rise. This will make their bodies more active. As a result, they will need more oxygen than they do now. Because the amount of oxygen will be (**31**), the fish will stop growing even earlier. As smaller fish tend to lay fewer eggs, this will lead to a drop in the overall population of the species. Even a tiny change in ocean temperatures will have a strong influence.

Global warming is very complicated, and it creates different problems depending on the area and the types of plants and animals found there. Because of this, (**32**) the results of it is very difficult. In an attempt to do this, scientists have recently created a computer program to calculate the impact that global warming will have on 600 different species of fish in the future. Using the program, they estimate that average body size will fall between 14 and 24 percent. Recent studies of two kinds of fish, however, suggest that this estimate is not accurate and the problem is even worse. Actually, the average body size of these fish has already been (**33**). These researchers say the change in size that they found is yet another reminder of the urgent need to do something to stop global warming.

- | | | | | |
|------|--------------|------------|--------------|--------------|
| (30) | 1 beneficial | 2 simple | 3 dramatic | 4 temporary |
| (31) | 1 limited | 2 generous | 3 typical | 4 divided |
| (32) | 1 repairing | 2 ignoring | 3 predicting | 4 publishing |
| (33) | 1 improved | 2 deleted | 3 reached | 4 affected |

The Road to Safety

Every year, millions of car accidents occur around the world. These accidents not only cause many deaths and injuries but also cost a great amount of money, both in medical expenses and in lost working days. Until recently, researchers have focused on finding ways to help people stay alive when these accidents happen. Now, however, thanks to the development of information technology, they have begun to take a different approach. This is to find ways to prevent traffic accidents from happening in the first place.

The United States government, together with a number of leading car manufacturers, is developing a system called V2X that does just this. In this system, each car is fitted with a device that sends out radio signals 10 times a second. These signals contain information about the car's speed, location, and direction. Each car also carries a receiver that picks up signals from other cars. Because of these signals, the device can detect the location of all other cars within 300 meters.

This system can be used to prevent many common kinds of accidents. For example, if a car ahead of you suddenly slows down or stops, the system will warn you to reduce your speed. Before you make a turn, it can tell you if another car has stopped around the corner. It can also tell you when it is safe to pass the car in front of you. Unlike earlier systems, which used cameras, this system can warn you about cars that you cannot actually see.

Experts believe that V2X will greatly reduce the number of traffic accidents. Some people, though, have concerns about it. One is that if the system gives out too many warnings, the driver will become annoyed or even lose concentration. Another concern is privacy. If drivers think that the system will be used to track where they go, they may refuse to use it. The U.S. Department of Transportation, however, insists that the information gained by the new system will only be used to help drivers drive more safely.

(37)

Until recently, researchers mainly tried to figure out

- 1 how to stop people from dying when they have been in car accidents.
- 2 why the majority of car accidents happen.
- 3 what the most common serious injuries caused by car accidents are.
- 4 when car accidents are most likely to occur.

(38)

The United States government and car manufacturers are developing a system that

- 1 uses radio signals to find cars that have been stolen by criminals.
- 2 requests information from drivers in different areas of the country.
- 3 will keep all cars at least 300 meters away from each other.
- 4 sends and receives signals to tell drivers where other cars are.

(39)

What is one difference between the V2X system and earlier systems?

- 1 It tells drivers where to turn if there is too much traffic caused by an accident.
- 2 It gives information to drivers about cars that they are not able to see.
- 3 It uses a special camera to help drivers see things that are around corners.
- 4 It shows drivers the locations of stop signs on the street where they are driving.

(40)

Why are some people worried about introducing V2X?

- 1 The system might make it difficult for drivers to concentrate.
- 2 The system might provide inaccurate driving information.
- 3 The system does not give warnings when cars are stopped.
- 4 The system has not been tested by the U.S. government.

Asteroid Mining

Many metals that are essential to modern technology, such as platinum and gold, are difficult to find and get out of the ground. In addition, the expansion of industry over the last few decades has led to a huge rise in demand for these metals. Prices have risen higher than ever before, and some experts think that many of these rare metals will be completely used up in around 60 years' time.

One possible solution to this problem that has been discussed for years is to try to get these metals from asteroids. Asteroids are large pieces of metal and rock that float in space and that vary in size from a few meters to several kilometers long. The main difficulty with this idea has been the great expense of sending rockets into space to find and mine these asteroids. Now, however, a new company called Planetary Resources, which has been founded by a group of American billionaires, claims it will be able to start removing valuable metals and other minerals from asteroids within 20 years.

The company says that the key to achieving this aim is to cut the cost of space travel by using new technology that has been developed over the last decade. The company says that it will first develop telescopes that can be sent up into space. These telescopes will be rented out to government space programs to make money for the project and also used by the company itself to find asteroids that could become a target for future mining. In the next stage, robot spacecraft will be sent to land on asteroids and find out what metals they contain. In the final stage, other spacecraft will be sent to mine them.

Some scientists doubt that the company will be able to reduce the cost of space travel enough to make a profit on the metals it finds. A spokesman for Planetary Resources, however, points out that a single 500-meter asteroid could contain more platinum than has been mined on Earth in its whole history. Although the company's plan sounds unrealistic to many people today, it has been founded by a group of successful businesspeople, and they, at least, believe that it will work.

(41) What is one reason the price of metals such as gold and platinum has gone up in recent years?

- 1 The new technology that is used to find them is extremely expensive.
- 2 There are fewer companies that find and sell them than before.
- 3 There has been a greater demand for them due to the growth of industry.
- 4 The amount of metals people can remove from the ground is controlled.

(42) So far, the main challenge of getting metals from asteroids has been that

- 1 the metals are deep inside the asteroids and difficult to reach.
- 2 the cost of sending rockets to find and remove the metals is too high.
- 3 the asteroids are hard to land on because they are always moving.
- 4 the number of people needed to mine an asteroid cannot fit on a rocket.

(43) How will Planetary Resources discover what kinds of metals an asteroid contains?

- 1 By paying countries with spacecraft to send people to asteroids.
- 2 By sending robot spacecraft to visit asteroids.
- 3 By asking governments for data on asteroids.
- 4 By breaking up asteroids using special telescopes placed in space.

(44) Why does Planetary Resources believe that it can make money by mining asteroids?

- 1 The amount of rare metals that one asteroid contains could be very large.
- 2 It will no longer have to pay for space travel in a few years.
- 3 Scientists have discovered a way to make the quality of the metals they find better.
- 4 Its plan seems so unrealistic to most people that no one else will try it first.

(45) Which of the following statements about Planetary Resources is true?

- 1 It says it can replace all of Earth's rare metals within the next 60 years.
- 2 It says it will be able to begin mining asteroids within the next 20 years.
- 3 It will buy used telescopes from government space agencies.
- 4 It can pay for its costs by finding billionaires to help the company.

3

A

次の英文 **A**, **B** を読み、その文意にそって(26) から(33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び、その番号を解答用紙の所定欄にマークしなさい。

Is It Worth It?

Every year, wealthy individuals and governments provide millions of dollars to help people in developing countries. Sometimes, though, this money is (**26**). One reason for this is that the people giving the money are often not familiar with the actual conditions in a local place. As a result, money is provided for projects that are not very useful. One example that has recently attracted attention is that of PlayPumps, an invention for getting water from the ground.

In some parts of Africa, sources of water are often far from where people live, and women and girls are usually responsible for walking long distances to get this water. This gives them less (**27**) to work or go to school. In the 1990s, Trevor Field, a South African businessman, met an inventor who had come up with a solution to this. The inventor had created a pump based on a common piece of playground equipment that spins in circles. Field and the inventor worked together to produce these pumps, called PlayPumps, which were then put into school playgrounds in South Africa. As the children played on the PlayPump, it pulled water up from deep within the ground. The children enjoyed it, and women and girls gained time to do other things.

The idea attracted wide support, and millions of dollars were given to install the pumps in schools in other areas of Africa. Unfortunately, though, various problems soon appeared. The pumps were expensive and also difficult to repair. (**28**), they often did not pump up enough water to cover needs. While they were useful in big towns, where many children played on them, they did not work well in little country villages. Now, many experts are saying that this is a good example of the need to allow local people to decide how to spend money. They usually know how to meet their own needs better than those who (**29**) it.

-
- | | | | | |
|------|----------------------|---------------------|-----------------------|---------------------|
| (26) | 1 made | 2 limited | 3 wasted | 4 taken |
| (27) | 1 opportunity | 2 permission | 3 necessity | 4 confidence |
| (28) | 1 In exchange | | 2 What is more | |
| | 3 Instead | | 4 However | |
| (29) | 1 regret | 2 research | 3 donate | 4 correct |

Jet Lag

As the world has become more globalized, the number of people traveling internationally has increased greatly. One problem that these people often encounter is “jet lag.” This is when a traveler’s body continues to follow the time of the country the traveler has left rather than the one he or she has arrived in. Jet lag makes it hard for people to sleep at the right time. (**30**), they often find it difficult to concentrate or work hard during the day.

The reason that people suffer from jet lag is that we all have a biological clock inside us. This clock tells our bodies when it is time to sleep or to wake up. It also controls other (**31**) functions. Blood pressure and body temperature, for example, rise and fall in cycles. Cycles such as these are controlled by the biological clock and are not directly affected by things happening outside our bodies. The biological clock itself, though, adjusts its timing to the environment. It mainly does this by (**32**) the current pattern of day and night. As it gets dark, the brain releases a chemical called melatonin. This chemical tells the body to go to sleep.

Because the biological clock responds to light levels, it eventually changes to fit the local time when we go to another country. This usually takes a long time, though, so scientists have been studying how the biological clock adjusts in hopes of finding ways to speed up the process. One method is to control the release of melatonin. Many scientists have found that it is easiest to do this by exposing the body to more light to make it adapt more rapidly. Another method is to use special medicine that acts just like melatonin in the brain. Scientists believe that, one day, using such (**33**) may make this problem a thing of the past.

- | | | |
|-------------|--|--|
| (30) | 1 By chance
3 For instance | 2 On the other hand
4 Due to this |
| (31) | 1 visual 2 physical | 3 artificial 4 cultural |
| (32) | 1 giving up
3 going against | 2 forgetting about
4 reacting to |
| (33) | 1 laws 2 customs | 3 clinics 4 techniques |

The New Kid in Class

Children often complain about going to school, but how would they feel if they could not go at all? In fact, there are children who cannot participate in school life because of illnesses, such as severe allergies. These children can be taught at home, but they cannot usually enjoy the experiences and friendships that most children have at school. Recently in the United States, a simple robot that can attend school in place of these children is being used to solve this problem.

The robot is called the VGo. Basically, it is a mobile communication device that can be controlled using a computer. A display screen shows the face of the child, and speakers allow his or her voice to be heard. A camera and microphones allow the child to see and hear what is happening at the school. The robot has wheels, which can be controlled over the Internet, letting the user move it around the school. In this way, the child can take part in school life much like any other child.

The VGo was originally developed for use by professionals in business and medicine. Business managers, for example, use it to visit distant factories and discuss how to improve working methods. Medical specialists also use the robots to meet with patients who are in small, rural hospitals that only have regular doctors. When a salesman for the robot visited a high school, though, the school staff there saw the possibility of using it in classrooms. They began using one to represent a child who could not leave home because of a disease that made his body unable to fight germs.

The VGo robot does have some disadvantages. One of these is its high price, which means volunteers often need to raise money to get one for a school. Another is that it cannot open doors or use stairs. This means the user often has to ask for help. Many schools, though, are happy to cooperate in finding solutions to these problems so that children who cannot attend school in person can still enjoy the life of a schoolchild.

(37)

What is one problem that some children have?

- 1 Their friends from school do not live close enough to them.
 - 2 Their schools no longer help them go to and from school.
 - 3 They cannot attend school because of their health.
 - 4 They are not allowed to be taught at home by their parents.
-

(38)

The VGo allows children to

- 1 see what is going on outside while they are at school.
 - 2 connect to the Internet and write their opinions online.
 - 3 take part in classes together with other students.
 - 4 move from room to room within their own homes.
-

(39)

What is the original reason that the VGo was developed?

- 1 To help professionals visit places that are far away.
 - 2 To allow patients to move around more easily in hospitals.
 - 3 To replace factory workers with robots that work faster.
 - 4 To let salesmen show people products that are too large to carry.
-

(40)

What is one problem with VGo robots?

- 1 It costs a lot to repair the damage they cause to doors and stairs.
- 2 Some younger children have trouble controlling them.
- 3 They can make it hard for some children to have a normal life.
- 4 Schools are often unable to pay for them without help.

Local Money

As business becomes more international, it becomes difficult for local economies to protect themselves against economic problems in other places. This is especially true when many different areas share the same type of currency, such as the euro in Europe. If there is a big economic problem in one place, this can make life difficult for everyone who uses the currency. For this reason, some local communities have recently started creating their own local money that people can use to buy goods and services in that area.

Because a local currency can only be used in one small area, the money stays there. This can help local businesses and create more jobs, giving people in the area more money to spend. It is often difficult for a local currency to succeed, though, because it cannot be used outside of that region and few stores in the community will accept it. One example of local money that has worked is the Chiemgauer, which was started by a high school teacher named Christian Gelleri in 2003 in the Rosenheim district of southern Germany.

Over 600 local businesses accept the Chiemgauer as payment, and it has become very popular. One reason the Chiemgauer has been successful is because it was well planned. Each Chiemgauer is worth one euro, but when people buy it, 3 percent of the money goes to local charities. This gives people another reason to support the money. Moreover, if businesses change the money back to euros, a fee of 5 percent is collected by the bank. This encourages companies to keep the Chiemgauer and use it to buy their supplies locally. This, in turn, supports the local economy.

At the moment, the Chiemgauer is only used in a small part of Rosenheim's economy. Gelleri believes that, eventually, the money will be used in about 50 percent of it. The currency will never completely replace the euro because big companies in Rosenheim need euros to purchase their supplies from other places. Using it will, though, make sure that people spend a higher percentage of their money locally, which will help to protect Rosenheim from big changes in the European economy.

(41)

Some areas are thinking about using their own local money

- 1 to avoid being affected by economic problems in other places.
- 2 to help the businesses in their own communities grow internationally.
- 3 because they want people visiting from elsewhere to buy more things.
- 4 because they have trouble selling their goods to countries in Europe.

(42)

What is one benefit of using local currencies?

- 1 It helps to promote school programs in the local area.
- 2 It can lead to the creation of more jobs in the community.
- 3 It teaches children in the community how money works.
- 4 It allows people to buy things at a discount from stores in the region.

(43)

When people buy the Chiemgauer,

- 1 a part of the money they pay is given to charities in the area.
- 2 the money they spend is used by the government to support the euro.
- 3 they are charged a fee of 5 percent that goes to local businesses.
- 4 they are given one week to spend their money in local shops.

(44)

What does Christian Gelleri think will happen in Rosenheim in the future?

- 1 Big businesses will switch from the euro to the local currency.
- 2 The economy will grow so much that it will not need a local money.
- 3 Local people will no longer need to buy supplies from other areas.
- 4 The Chiemgauer will be used in about half of the local economy.

(45)

Which of the following statements is true?

- 1 Using a currency such as the euro is important for protecting jobs.
- 2 Banks will not trade local money for major currencies.
- 3 Local currencies are often not accepted by businesses in a region.
- 4 The Chiemgauer is mainly used at German high schools.